DIB isolators, EEPROM, ADC, MCU

File: DIB_EEPROM_ADC_MCU.kicad_sch

DAC, OE, Vref, IOEXP, Temp

File: DAC_Vref_IOEXP_Temp.kicad_sch

Power pre-regulator

File: Power_pre-regulator.kicad_sch

Bias power

File: Bias_power.kicad_sch

Post-regulator

File: Post-regulator.kicad_sch

Rsense protection, power output, Rprog

File: Rsense_protection_power_output.kicad_sch

Mode detection, OVP

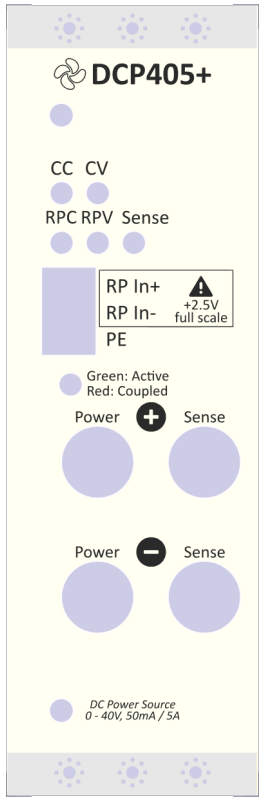
File: CC_CV_OVP.kicad_sch

Front mask details:

Material: Aluminum PCB
Thickness: 1.6 mm
PCB layout: 1-layer

DCP405+ project repository: https://github.com/eez-open/dib-dcp405plus/blob/main/KiCad/EEZ%20DIB%20DCP405plus%20front%20mask.kicad_pro
DCP405B+ project repository: https://github.com/eez-open/dib-dcp405plus/blob/main/KiCad/EEZ%20DIB%20DCP405plus%20front%20mask%20(Basic).kicad_pro





KK1 PCB-MB-01

KK2 PCB-MB-01

KK3 M3X8/D7985B

KK4 M3X8/D7985B

KK5 M3X8/D7985B

KK6 M3X8/D7985B

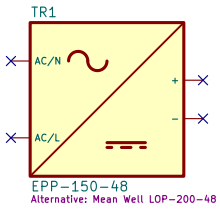
KK10 B3/BN2842

KK7 M3X8/D7985B

KK8 SK96/84SA

KK9 SP400-0.007-00-104

M1 TBP03P2-350-03BEOR



KK11 M3X4/D7985

KK12 M3X4/D7985

KK13 M3X4/D7985

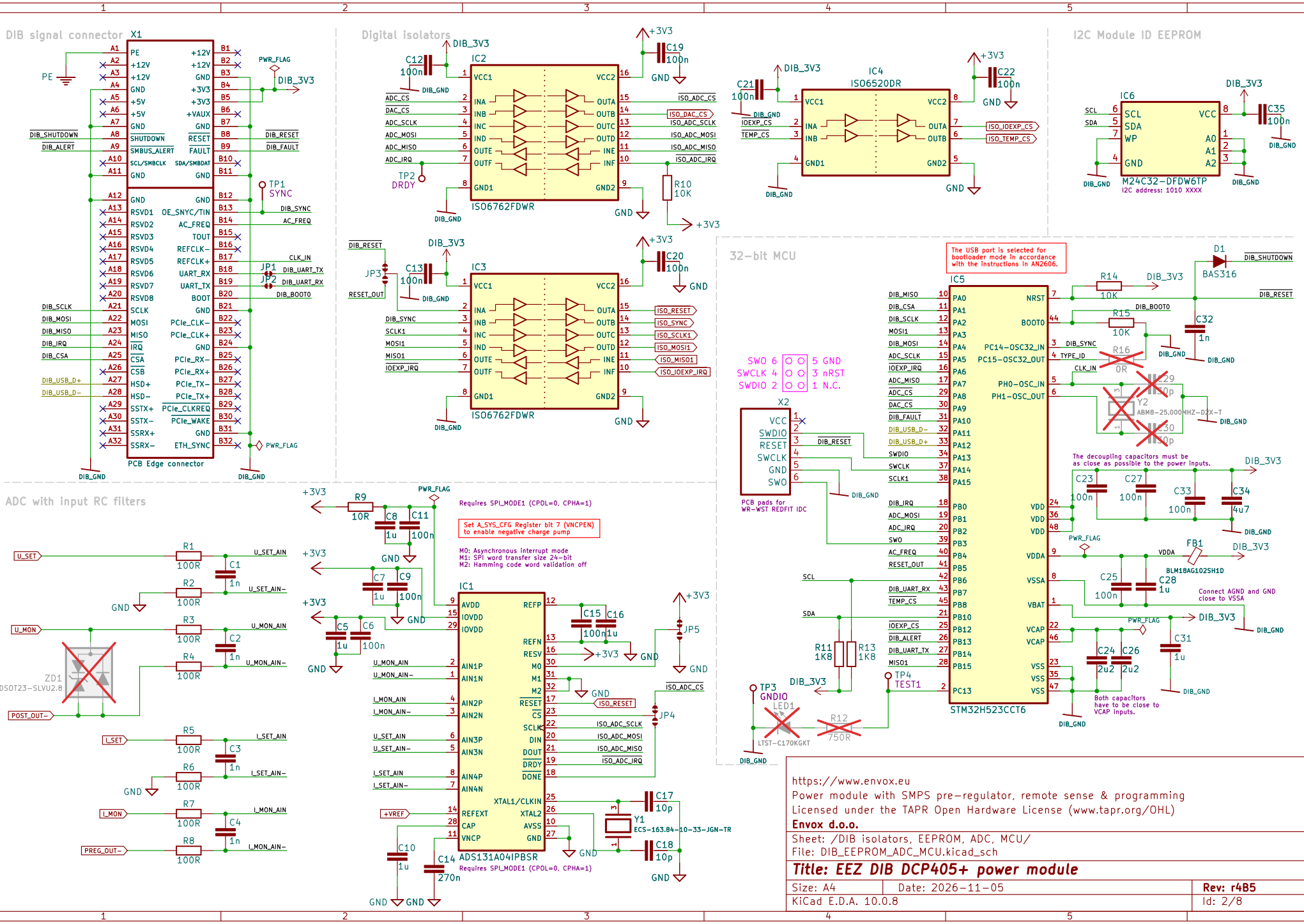
KK14 M3X4/D7985

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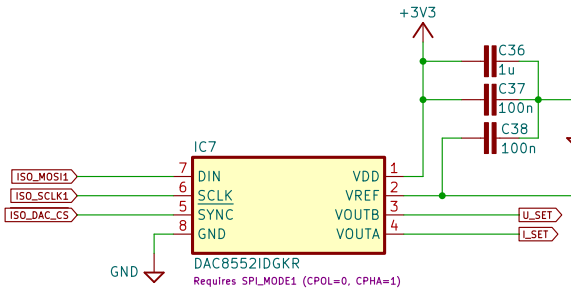
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File: EEZ DIB DCP405plus.kicad_sch

Title: EEZ DIB DCP405+ power module

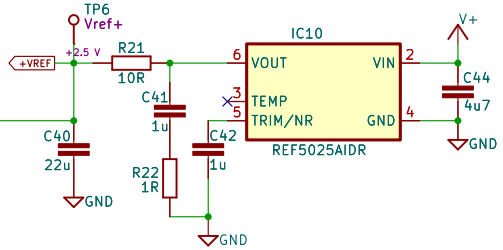
Size: A4	Date: 2026-11-05	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 1/8


DIB signal connector
X1
PCB Edge connector
Digital isolators
IC2
ISO6762FDWR
IC3
ISO6762FDWR
IC4
ISO6520DR
I2C Module ID EEPROM
IC6
M24C32-DFDW6TP
I2C address: 1010 XXXX
32-bit MCU
IC5
STM32H523CCT6
The USB port is selected for bootloader mode in accordance with the instructions in AN2606.
PCB pads for WR-WST REDFIT IDC
The decoupling capacitors must be as close as possible to the power inputs.
Connect AGND and GND close to VSSA
Both capacitors have to be close to VCAP inputs.
ADC with input RC filters
IC1
ADS131A04IPBSR
Requires SPI_MODE1 (CPOL=0, CPHA=1)
Set A_SYS_CFG Register bit 7 (VNCPEN) to enable negative charge pump
M0: Asynchronous interrupt mode
M1: SPI word transfer size 24-bit
M2: Hamming code word validation off
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Sheet: /DIB isolators, EEPROM, ADC, MCU/
File: DIB_EEPROM_ADC_MCU.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-05
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 2/8

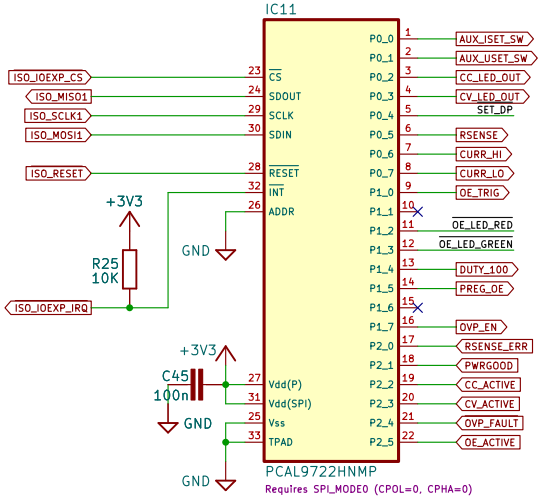
16-bit DAC (set voltage and current)



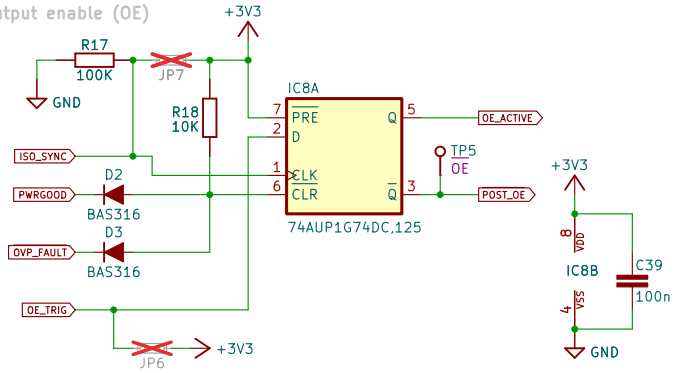
Voltage reference



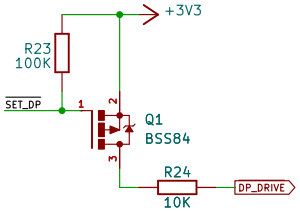
SPI 22-bit I/O Expander



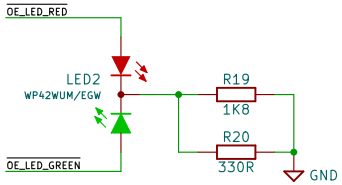
Output enable (OE)



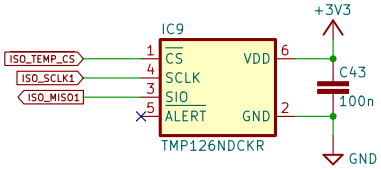
Down-programmer



OE indicator



SPI Temperature sensor

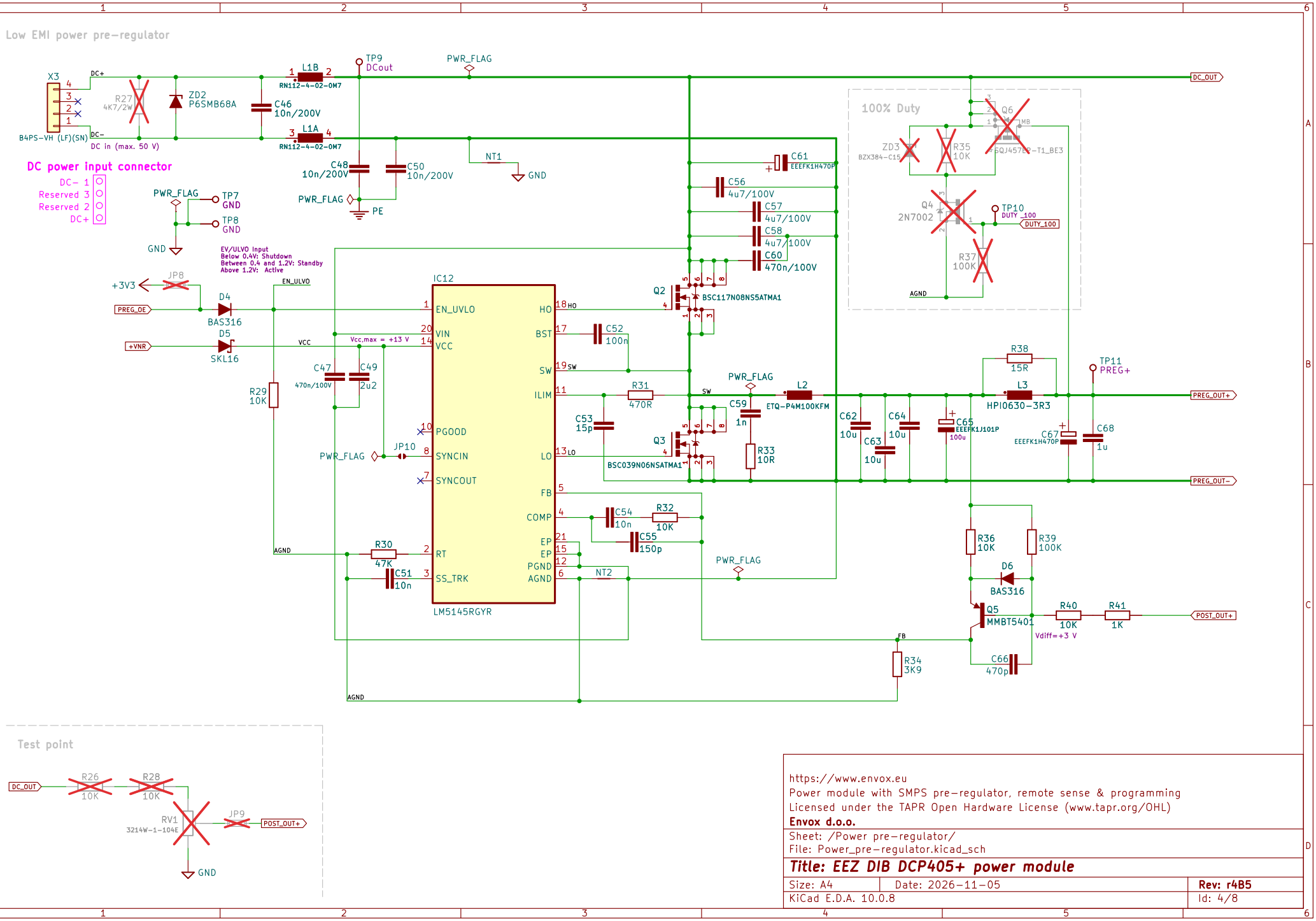


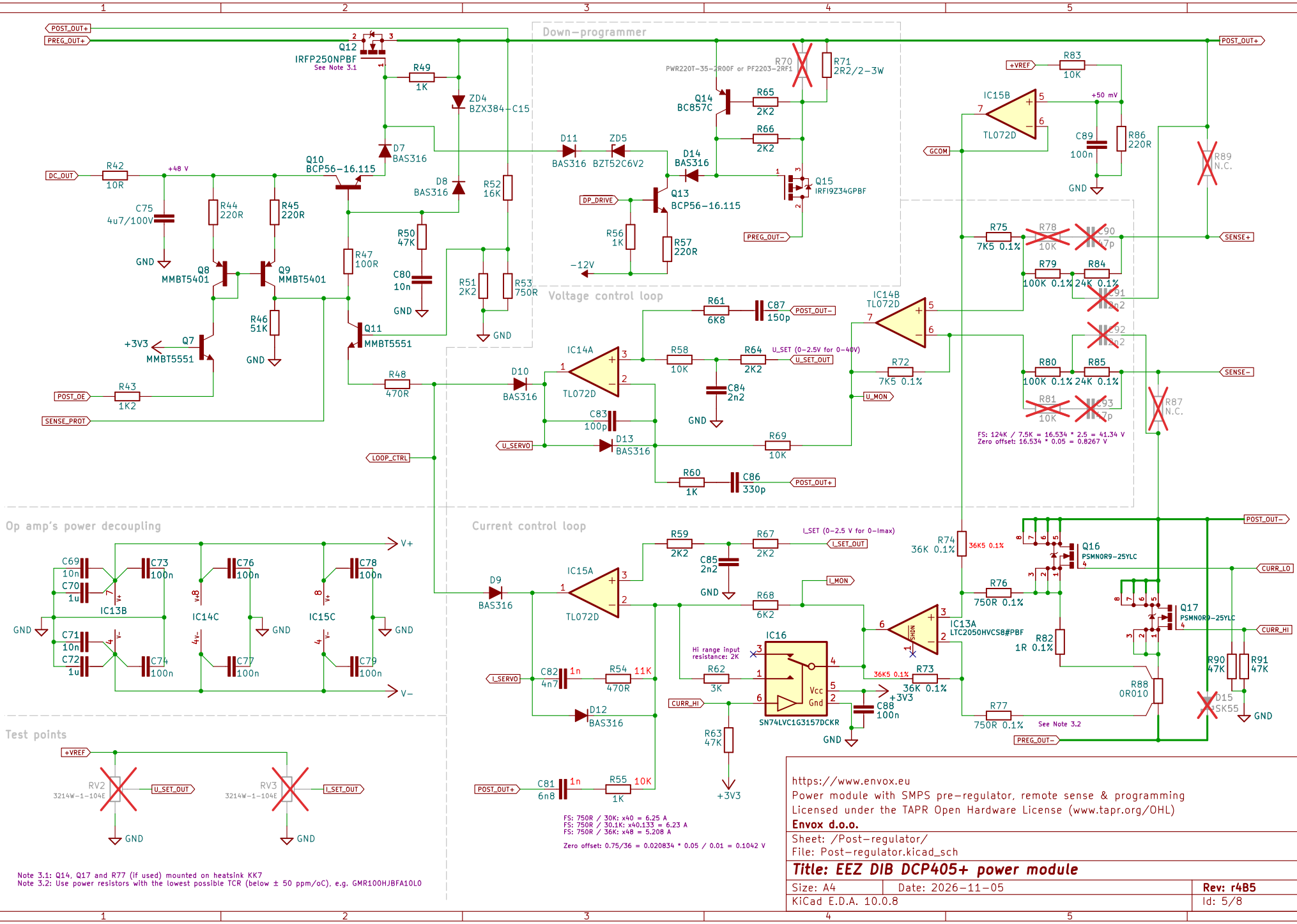
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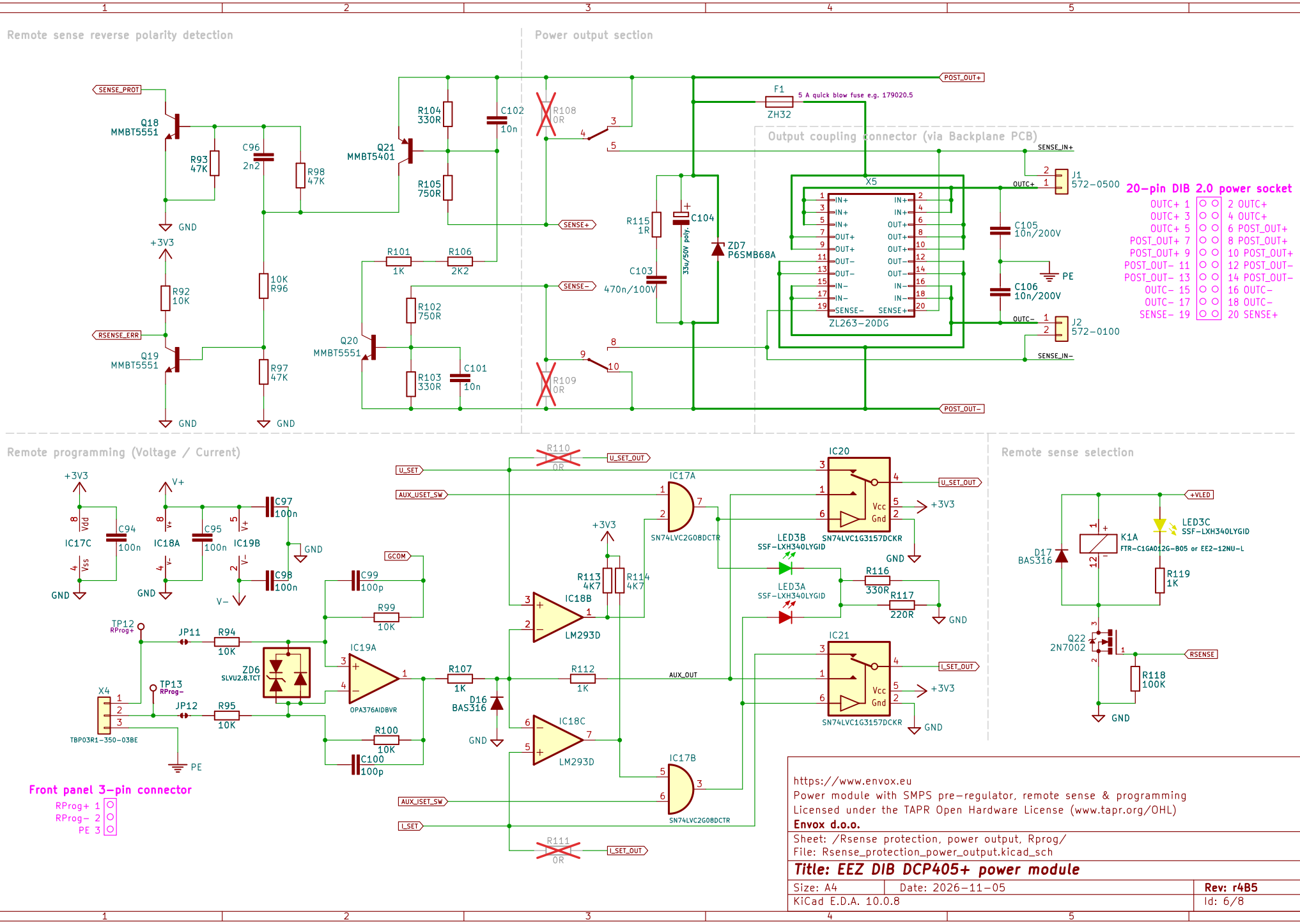
Sheet: /DAC, OE, Vref, IOEXP, Temp/
File: DAC_Vref_IOEXP_Temp.kicad_sch

Title: EEZ DIB DCP405+ power module

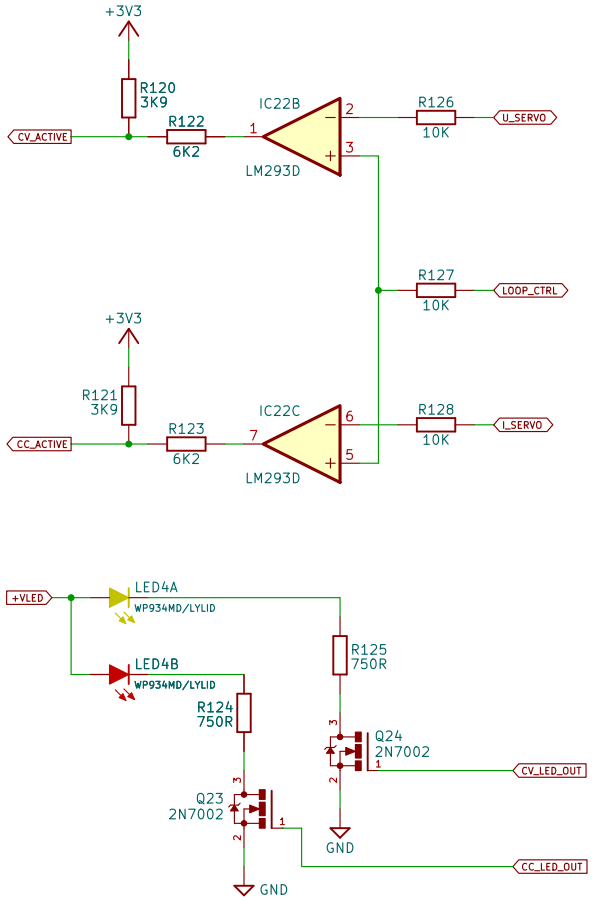
Size: A4	Date: 2026-11-05	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 3/8


Low EMI power pre-regulator
DC power input connector
DC- 1
Reserved 3
Reserved 2
DC+
X3
B4PS-VH (LF)(SN)
DC in (max. 50 V)
R27 4K7/2W
ZD2 P6SMB68A
C46 10n/200V
L1B RN112-4-02-0M7
L1A RN112-4-02-0M7
TP9 DCout
PWR_FLAG
C48 10n/200V
C50 10n/200V
PE
NT1
GND
TP7 GND
TP8 GND
JP8
+3V3
PREG_OE
D4 BAS316
+VNR
D5 SKL16
EN_UVLO
VCC
Vcc,max = +13 V
EV/UVLO Input
Below 0.4V: Shutdown
Between 0.4 and 1.2V: Standby
Above 1.2V: Active
C47 470n/100V
C49 2u2
R29 10K
IC12
EN_UVLO
VIN
VCC
PGOOD
SYNCIN
SYNCOUT
RT
SS_TRK
HO
BST
SW
ILIM
LO
FB
COMP
EP
EP
PGND
AGND
LM5145RGYR
JP10
R30 47K
C51 10n
AGND
C52 100n
R31 470R
C53 15p
C54 10n
R32 10K
C55 150p
NT2
Q2 BSC117N08NS5ATMA1
Q3 BSC039N06NSATMA1
C61 EEEFK1H470P
C56 4u7/100V
C57 4u7/100V
C58 4u7/100V
C60 470n/100V
C59 1n
R33 10R
L2 ETQ-P4M100KFM
SW
100% Duty
ZD3 BZX384-C15
R35 10K
Q6 SQJ457EP-T1_BE3
Q4 2N7002
TP10 DUTY_100
DUTY_100
R37 100K
AGND
R38 15R
L3 HPI0630-3R3
TP11 PREG+
C62 10u
C64 10u
C63 10u
C65 EEEFK1J101P 100u
C67 EEEFK1H470P
C68 1u
PREG_OUT+
PREG_OUT-
R36 10K
R39 100K
D6 BAS316
Q5 MMBT5401
R40 10K
R41 1K
Vdiff=+3 V
POST_OUT+
C66 470p
FB
R34 3K9
Test point
DC_OUT
R26 10K
R28 10K
RV1 3214W-1-104E
JP9
POST_OUT+
GND
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Sheet: /Power pre-regulator/
File: Power_pre-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-05
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 4/8

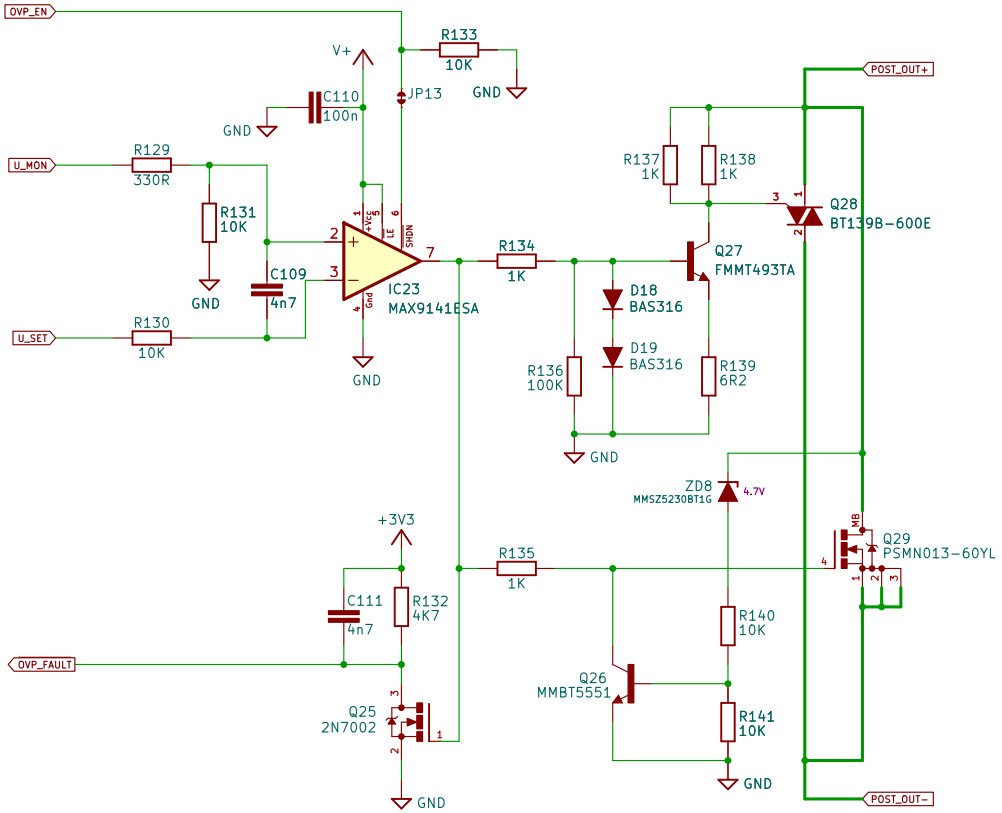

Down-programmer
Voltage control loop
Current control loop
Op amp's power decoupling
Test points
POST_OUT+
PREG_OUT+
DC_OUT
+48 V
Q12
IRFP250NPBF
See Note 3.1
R49
1K
ZD4
BZX384-C15
D7
BAS316
Q10
BCP56-16.115
D8
BAS316
R42
10R
C75
4u7/100V
R44
220R
R45
220R
R47
100R
R50
47K
C80
10n
R52
16K
R51
2K2
R53
750R
GND
Q8
MMBT5401
Q9
MMBT5401
Q7
MMBT5551
Q11
MMBT5551
+3V3
R46
51K
R43
1K2
POST_OE
SENSE_PROT
R48
470R
LOOP_CTRL
D11
BAS316
ZD5
BZT52C6V2
D14
BAS316
DP_DRIVE
R56
1K
Q13
BCP56-16.115
R57
220R
-12V
PREG_OUT-
R70
PWR220T-35-2R00F or PF2203-2RF1
R71
2R2/2-3W
Q14
BC857C
R65
2K2
R66
2K2
Q15
IRFI9Z34GPBF
R61
6K8
C87
150p
POST_OUT-
IC14A
TL072D
R58
10K
R64
2K2
U_SET (0-2.5V for 0-40V)
U_SET_OUT
C84
2n2
D10
BAS316
C83
100p
U_SERVO
D13
BAS316
R69
10K
R60
1K
C86
330p
IC14B
TL072D
R72
7K5 0.1%
U_MON
+VREF
R83
10K
IC15B
TL072D
+50 mV
C89
100n
R86
220R
GCOM
R89
N.C.
R75
7K5 0.1%
R78
10K
C90
47p
R79
100K 0.1%
R84
24K 0.1%
C91
2n2
C92
2n2
R80
100K 0.1%
R85
24K 0.1%
R81
10K
C93
47p
R87
N.C.
SENSE+
SENSE-
FS: 124K / 7.5K = 16.534 * 2.5 = 41.34 V
Zero offset: 16.534 * 0.05 = 0.8267 V
C69
10n
C70
1u
IC13B
C73
100n
C71
10n
C72
1u
C74
100n
IC14C
C76
100n
C77
100n
IC15C
C78
100n
C79
100n
V+
V-
R59
2K2
C85
2n2
D9
BAS316
IC15A
TL072D
R67
2K2
I_SET (0-2.5 V for 0-Imax)
I_SET_OUT
I_MON
R68
6K2
IC13A
LTC2050HVCS8#PBF
IC16
Hi range input resistance: 2K
R62
3K
SN74LVC1G3157DCKR
Vcc
Gnd
+3V3
R63
47K
C88
100n
R73
36K 0.1%
R74
36K 0.1%
R76
750R 0.1%
R82
1R 0.1%
R77
750R 0.1%
See Note 3.2
Q16
PSMN0R9-25YLC
Q17
PSMN0R9-25YLC
CURR_LO
CURR_HI
R88
0R010
R90
47K
R91
47K
D15
PSK55
I_SERVO
C82
4n7
1n
R54
470R
11K
D12
BAS316
C81
6n8
1n
R55
1K
10K
FS: 750R / 30K: x40 = 6.25 A
FS: 750R / 30.1K: x40.133 = 6.23 A
FS: 750R / 36K: x48 = 5.208 A
Zero offset: 0.75/36 = 0.020834 * 0.05 / 0.01 = 0.1042 V
RV2
3214W-1-104E
RV3
3214W-1-104E
Note 3.1: Q14, Q17 and R77 (if used) mounted on heatsink KK7
Note 3.2: Use power resistors with the lowest possible TCR (below ± 50 ppm/oC), e.g. GMR100HJBFA10L0
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Sheet: /Post-regulator/
File: Post-regulator.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-05
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 5/8


Remote sense reverse polarity detection
Power output section
Output coupling connector (via Backplane PCB)
Remote programming (Voltage / Current)
Remote sense selection
Front panel 3-pin connector
RProg+ 1
RProg- 2
PE 3
20-pin DIB 2.0 power socket
OUTC+ 1
2 OUTC+
OUTC+ 3
4 OUTC+
OUTC+ 5
6 POST_OUT+
POST_OUT+ 7
8 POST_OUT+
POST_OUT+ 9
10 POST_OUT+
POST_OUT- 11
12 POST_OUT-
POST_OUT- 13
14 POST_OUT-
OUTC- 15
16 OUTC-
OUTC- 17
18 OUTC-
SENSE- 19
20 SENSE+
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Sheet: /Rsense protection, power output, Rprog/
File: Rsense_protection_power_output.kicad_sch
Title: EEZ DIB DCP405+ power module
Size: A4
Date: 2026-11-05
Rev: r4B5
KiCad E.D.A. 10.0.8
Id: 6/8

Constant Voltage (CV) and Constant Current (CC) mode detection and indication



Over-voltage protection (OVP)



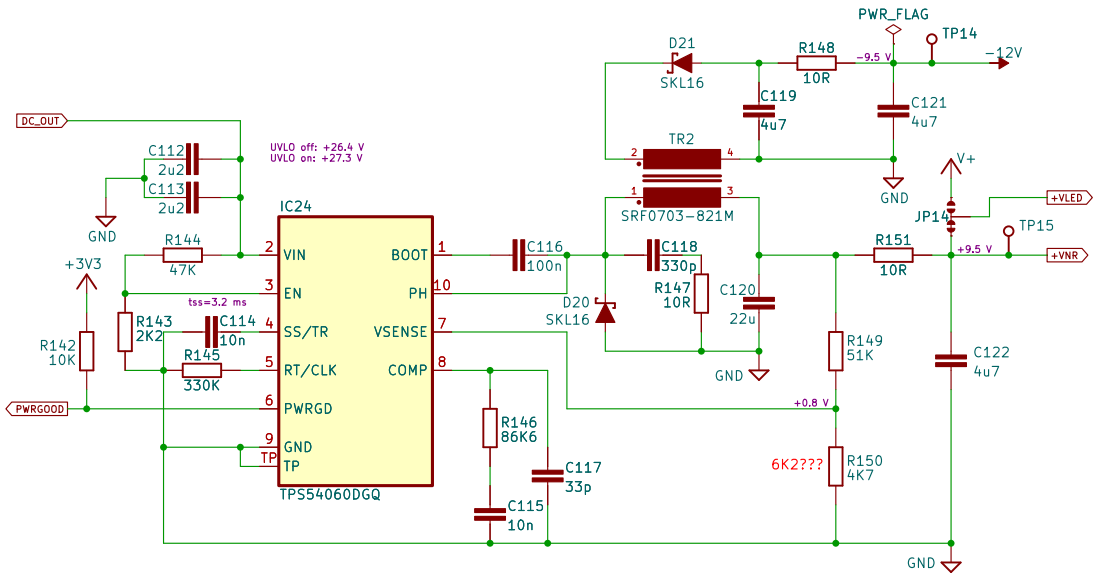
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Sheet: /Mode detection, OVP/
File: CC_CV_OVP.kicad_sch

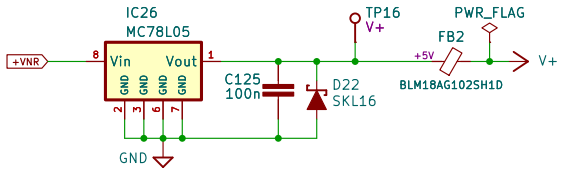
Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-11-05	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 7/8

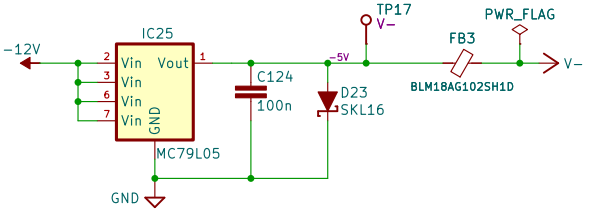
Bias power supply



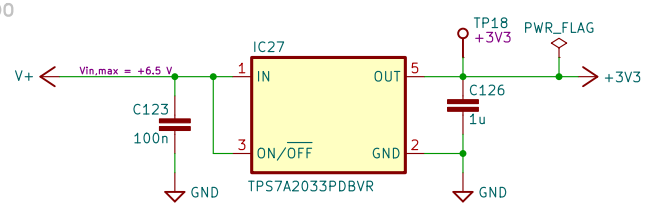
+5V LDO



-5V LDO



+3V3 LDO



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Sheet: /Bias power/
File: Bias_power.kicad_sch

Title: EEZ DIB DCP405+ power module

Size: A4	Date: 2026-11-05	**Rev: r4B5**
KiCad E.D.A. 10.0.8		Id: 8/8